

Notes: Kale and Shahrur (JFE, 2007)

Corporate Capital Structure and the Characteristics of Suppliers and Customers

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1 Big picture

- **Research question.** Do firms choose capital structure partly in response to the characteristics of their suppliers and customers? More specifically, does leverage reflect the need to preserve relationship-specific investments and the desire to bargain with powerful trading partners?
- **Main idea.** Capital structure is not only about taxes, bankruptcy costs, asymmetric information, and agency conflicts inside the firm. It can also shape implicit contracts with stakeholders outside the firm, especially suppliers and customers.
- **Relationship-specific investment channel.** If suppliers or customers must make investments that lose value when the firm fails or liquidates, the firm can credibly reassure them by choosing lower leverage. Lower debt reduces expected financial distress and makes the firm more willing to preserve reputation and ongoing relationships.
- **Bargaining channel.** Debt can also be useful as a bargaining device. When suppliers or customers are concentrated and therefore powerful, higher leverage can reduce the surplus available for them to extract, improving the firm's bargaining position.
- **Empirical object.** The paper relates firm leverage to weighted characteristics of supplier and customer industries, and then repeats the analysis using firm-level data on key suppliers and key customers.
- **Main finding.** Firm leverage is negatively related to supplier and customer R&D intensity and to supplier/customer strategic alliance and joint venture intensity. Firm leverage is positively related to concentration in supplier and customer industries.
- **Economic takeaway.** Capital structure helps manage relationships along the vertical chain. Firms use less debt when debt threatens relationship-specific investment, but more debt when debt helps bargain with concentrated trading partners.

2 Incremental contribution of the paper

- **Moves capital structure beyond firm boundaries.** The paper extends the implicit-contracting view of capital structure by treating suppliers and customers as important stakeholders in leverage choice.
- **Separates two opposite product-market motives for debt.** One motive predicts lower leverage: commitment to suppliers/customers that make relationship-specific investments. The other predicts higher leverage: bargaining against powerful suppliers/customers.
- **Builds supplier/customer measures from input-output data.** The paper uses U.S. benchmark input-output tables to identify each industry pair's supplier and customer links and then constructs weighted measures of supplier/customer R&D, concentration, and growth.
- **Uses both industry-level and firm-level links.** The industry-level sample is large and less exposed to individual-firm reverse causality, while the firm-level key supplier/customer sample more precisely identifies actual trading partners.
- **Introduces alliance and joint venture intensity as an additional relationship-specific investment proxy.** The idea is that vertical alliances and JVs are more common when repeated cooperation and relationship-specific investments are important.

- **Documents economically large effects.** The estimated effects of supplier/customer variables on leverage are comparable to traditional capital-structure determinants such as firm size, collateral value, profitability, Tobin’s q, R&D intensity, and selling-expense intensity.
- **Connects vertical integration to leverage.** If vertical integration already mitigates relationship-specific contracting problems, then leverage should be less important as a commitment mechanism. The evidence is consistent with this logic.

3 Data and measurement

3.1 Firm sample and period

The starting point is all Compustat firms from 1984 to 2003. The paper excludes financial firms, utilities, foreign-headquartered firms, and, for some tests, retail and wholesale firms. The resulting samples differ across supplier, customer, and key trading-partner analyses because not every firm has identifiable suppliers or customers.

- **Supplier industry-level sample.** 76,290 firm-year observations for 10,310 firms.
- **Customer industry-level sample.** 26,300 firm-year observations for 4,065 firms. Firms are included if no more than 25 percent of their industry’s output is sold to final users.
- **Key supplier firm-level sample.** 4,122 customer-year observations where key suppliers can be identified.
- **Key customer firm-level sample.** 9,452 supplier-year observations where key customers can be identified.

3.2 Leverage measures

The paper uses two leverage variables:

- **Market leverage:** book debt divided by book debt plus market value of common equity.
- **Book leverage:** book debt divided by book value of assets.

The main regressions use market leverage; book leverage is used as a robustness check.

3.3 Relationship-specific investment variables

The paper uses two types of proxies for relationship-specific investments.

R&D-based proxies. Supplier and customer R&D intensities are weighted averages of the R&D intensities of upstream and downstream industries. For a firm in industry i ,

$$\text{Supplier Industries R\&D}_i = \sum_{j \neq i} \text{Supplier Industry R\&D}_j \times \text{Industry Input Coefficient}_{ji},$$

$$\text{Customer Industries R\&D}_i = \sum_{j \neq i} \text{Customer Industry R\&D}_j \times \text{Industry Percentage Sold}_{ji}.$$

High supplier R&D means the firm relies on inputs from R&D-intensive supplier industries. High customer R&D means the firm’s output is sold to R&D-intensive customer industries.

Strategic alliance and joint venture proxies. Supplier or customer SA/JV intensity is the number of strategic alliances and joint ventures formed with supplier or customer industries, divided by the number of firms in the sample firm’s industry. The logic is that alliances and JVs are more common when vertical cooperation and relationship-specific investments are important.

3.4 Bargaining power variables

Supplier and customer concentration are weighted averages of Herfindahl indices in supplier and customer industries:

$$\begin{aligned}\text{Supplier Concentration}_i &= \sum_{j \neq i} \text{Herfindahl}_j \times \text{Industry Input Coefficient}_{ji}, \\ \text{Customer Concentration}_i &= \sum_{j \neq i} \text{Herfindahl}_j \times \text{Industry Percentage Sold}_{ji}.\end{aligned}$$

Higher supplier concentration means the firm’s input providers have greater seller power. Higher customer concentration means buyers have greater buyer power.

3.5 Controls and sample cleaning

The regressions control for standard capital-structure determinants: firm size, asset collateral value, volatility, nondebt tax shields, return on assets, Tobin’s q, own R&D intensity, selling-expense intensity, industry concentration, market share, and industry/year fixed effects. Variables are winsorized at the 1st and 99th percentiles.

4 Methodology and empirical specifications

4.1 Hypothesis 1: relationship-specific investments

The central commitment hypothesis is:

Firms that deal with R&D-intensive suppliers/customers, or firms with high alliance and joint venture intensity with suppliers/customers, choose lower leverage.

A generic industry-level regression is:

$$\begin{aligned}\text{Leverage}_{it} &= \alpha + \beta_1 \text{Supplier/Customer R\&D}_{it} + \beta_2 \text{Supplier/Customer SA/JV}_{it} \\ &\quad + \beta_3 \text{Supplier/Customer Concentration}_{it} + \Gamma' X_{it-1} + \text{Industry FE} + \text{Year FE} + \varepsilon_{it}.\end{aligned}$$

The expected signs under Hypothesis 1 are $\beta_1 < 0$ and $\beta_2 < 0$.

4.2 Vertical integration tests

If vertical integration solves the relationship-specific investment problem, then leverage should matter less as a commitment device for vertically integrated firms.

- **Backward integration dummy:** equals one if the firm has a segment in a supplier industry.
- **Forward integration dummy:** equals one if the firm has a segment in a customer industry.

The paper interacts these dummies with supplier/customer R&D or SA/JV measures. A positive interaction with the R&D variable weakens the negative baseline relation between R&D and leverage.

4.3 Market share and importance to customers

The effect of customer relationship-specific investment should be stronger when the firm is important to its customers. The paper uses the firm’s market share and industry concentration as proxies for the scarcity of alternative suppliers. Negative interactions between customer R&D and market share, or between customer R&D and industry concentration, support this idea.

4.4 Hypothesis 2: debt as a bargaining tool

The bargaining hypothesis is:

A firm’s leverage is positively related to concentration in its supplier and customer industries.

The expected coefficient on supplier/customer concentration is positive. If the firm itself has high market share, it already has bargaining power, so the positive relation between trading-partner concentration and debt should be weaker. This predicts a negative interaction between supplier/customer concentration and the firm’s market share.

4.5 Firm-level key supplier and customer tests

The firm-level analysis identifies key customers using Compustat segment disclosures and matches customers and suppliers manually and algorithmically. It then constructs:

- **Key Suppliers R&D:** weighted R&D intensity of key supplier firms.
- **Key Customers R&D:** weighted R&D intensity of key customer firms.
- **Supplier/Customer SA and JV Dummy:** equals one if the firm has at least one alliance or JV with a key supplier/customer.

These tests are more precise about actual trading partners but are also more vulnerable to joint determination of leverage and R&D.

4.6 Simultaneous equation models

To address joint determination, the paper estimates 2SLS systems. For supplier R&D and customer leverage, the system is:

$$\begin{aligned}\text{Customer Market Leverage} &= \beta_0 + \beta_1 \text{Supplier R\&D} + \Gamma'X + \varepsilon, \\ \text{Supplier R\&D} &= \delta_0 + \delta_1 \text{Customer Market Leverage} + H'Y + \mu.\end{aligned}$$

The analogous customer system relates supplier leverage to customer R&D. The key empirical prediction is that the leverage equation and the trading-partner R&D equation both show negative relations.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies robust conditional associations between leverage and supplier/customer characteristics. The strongest tests use industry-level variables because a single firm's leverage is unlikely to determine the R&D intensity, concentration, or alliance intensity of entire supplier and customer industries.

5.2 Why R&D proxies relationship-specific investment

R&D-intensive vertical chains often involve specialized inputs, technological interdependence, and assets that lose value outside a particular relationship. R&D is therefore a plausible proxy for relationship-specific investment, especially when both sides are innovation-intensive.

5.3 Why alliances and joint ventures are useful proxies

Strategic alliances and JVs are observable signs of close vertical cooperation. If supplier/customer relationships require relationship-specific investment, firms may formalize cooperation through alliances or JVs. The paper finds lower leverage when such vertical cooperation is more common.

5.4 Endogeneity concerns

The paper addresses endogeneity in several ways:

- uses industry-level supplier/customer characteristics, which are less likely to be caused by a single firm's leverage;
- includes many standard firm controls and industry/year dummies;
- includes supplier/customer growth controls so R&D variables do not merely proxy for upstream or downstream growth opportunities;
- estimates between-firm and Fama–MacBeth-style specifications;
- uses firm-level simultaneous equation models for key suppliers and customers.

5.5 Interpretation of the negative R&D coefficient

A negative coefficient on supplier or customer R&D does not mean that R&D mechanically causes lower debt in every setting. It means that firms exposed to R&D-intensive trading partners choose lower leverage after controlling for firm characteristics and industry/year effects. The interpretation is that low leverage helps preserve implicit contracts and encourages relationship-specific investments.

5.6 Interpretation of the positive concentration coefficient

A positive coefficient on supplier or customer concentration supports the bargaining role of debt. When suppliers or customers are more concentrated, they have more bargaining power. Debt can credibly reduce the surplus available for extraction, so firms facing powerful trading partners choose higher leverage.

5.7 Limits of the evidence

The design is not a randomized experiment. R&D and alliances are proxies, not direct measures of relationship-specific investment. Input-output tables measure industry-level vertical links, not every actual transaction. Firm-level key supplier/customer data are more precise but smaller and more endogenous. The paper is best read as strong evidence that vertical-chain relationships are important omitted determinants of leverage.

6 Tables and figures

6.1 Tables 1 and 2: Descriptive statistics and univariate evidence

Read Table 1:

- The mean market leverage in the overall sample is 0.187 and the mean book leverage is 0.268.
- The supplier industry-level sample has 76,290 observations, while the customer industry-level sample has 26,300 observations.
- The key supplier and key customer firm-level samples are smaller, with 4,122 and 9,452 observations, respectively.
- Supplier and customer variables show substantial variation, which makes the vertical-chain tests feasible.

Read Table 2:

- Firms facing above-median supplier R&D have much lower market leverage than firms facing below-median supplier R&D.
- Firms with higher supplier/customer SA and JV intensity also have lower leverage.
- Firms facing above-median supplier or customer concentration generally have higher market leverage, consistent with the bargaining hypothesis.

Economic message. The raw data already line up with the two main theories: relationship-specific investment variables are associated with lower leverage, while bargaining-power variables are associated with higher leverage.

Table 1

Descriptive statistics

This table reports descriptive statistics for the dependent and independent variables. The sample period is from 1984 to 2003. *Market Leverage* is the book value of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. *Book Leverage* is the book value of long-term debt and debt in current liabilities divided by the book value of assets. *Supplier Industries R&D* (*Customer Industries R&D*) is the weighted average of the R&D intensities of all supplier (customer) industries. *Supplier (Customer) SA and JV Intensity* is the number of SAs and JVs established between industry firms and firms in supplier (customer) industries, divided by the number of firms in the industry. *Compensation of Employees* is total compensation paid to employees divided by total output, constructed at the industry level. *Supplier Concentration* (*Customer Concentration*) is the weighted average of the Herfindahl indices of all supplier (customer) industries. *Supplier (Customer) Change in Sales* is a weighted average of the change in sales for supplier (customer) industries. *Key Suppliers R&D* (*Key Customers R&D*) is the weighted average of the R&D intensities of key suppliers (key customers). *Suppliers(Customer) SA and JV Dummy* is a dummy variable that equals one if the firm has at least one alliance or a SA established with one of its suppliers (customers), and zero otherwise. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles.

Variable	Mean	Median	Maximum	Minimum	Std dev.
Market Leverage (overall sample)	0.187	0.132	0.750	0.000	0.189
Book Leverage (overall sample)	0.268	0.210	1.882	0.000	0.290
<i>Supplier Industry-Level Variables (76,290 obs.)</i>					
Supplier Industries R&D	0.009	0.006	0.040	0.000	0.007
Supplier SA and JV Intensity	0.486	0.218	5.166	0.000	0.795
Compensation of Employees	0.303	0.299	0.818	0.022	0.116
Supplier Concentration	0.090	0.085	0.250	0.029	0.039
Supplier Change in Sales	0.098	0.096	0.308	-0.050	0.066
<i>Customer Industry-Level Variables (26,300 obs.)</i>					
Customer Industries R&D	0.017	0.009	0.081	0.000	0.017
Customer SA and JV Intensity	0.493	0.176	6.469	0.000	0.910
Customer Concentration	0.167	0.167	0.321	0.068	0.058
Customer Change in Sales	0.081	0.082	0.217	-0.030	0.047
<i>Supplier Firm-Level Variables(4,122 obs.)</i>					
Key Suppliers R&D	0.002	0.000	0.059	0.000	0.008
Supplier SA and JV Dummy	0.283	0.000	1.000	0.000	0.450
<i>Customer Firm-Level Variables(9,452 obs.)</i>					
Key Customers R&D	0.015	0.007	0.104	0.000	0.021
Customer SA and JV Dummy	0.162	0.000	1.000	0.000	0.368
<i>Control Variables(overall sample)</i>					
Industry Concentration	0.223	0.179	0.865	0.043	0.163
Firm Size (log of total assets)	4.280	4.146	10.115	-1.139	2.188
Asset Collateral Value	0.294	0.235	0.917	0.001	0.226
Volatility	0.201	0.091	2.963	0.010	0.381
Nondebt Tax Shield	0.001	0.000	0.017	0.000	0.002
Return on Assets	0.015	0.104	0.439	-2.284	0.344
Tobin's q	2.283	1.459	18.561	0.528	2.558
R&D Intensity	0.058	0.005	0.732	0.000	0.115
SE Intensity	0.332	0.250	2.374	0.000	0.354

Table 1: Descriptive statistics.

Table 2

Univariate analysis

This table reports univariate analysis for the difference in means of the leverage variables for different subsamples formed based on the median values of the supplier and customer variables. The sample period is from 1984 to 2003. *Market Leverage* is sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. *Book Leverage* is the book value of long-term debt and debt in current liabilities divided by the book value of assets. *Supplier Industries R&D* (*Customer Industries R&D*) is the weighted average of the R&D intensities of all supplier (customer) industries. *Key Suppliers R&D* (*Key Customers R&D*) is the weighted average of the R&D intensities of key suppliers (key customers). *Supplier (Customer) SA and JV Intensity* is the number of SAs and JVs established between industry firms and firms in supplier (customer) industries, divided by the number of firms in the industry. *Suppliers(Customer) SA and JV Dummy* is a dummy variable that equals one if the firm has at least one alliance or a SA established with one of its suppliers (customers), and zero otherwise. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier Concentration* (*Customer Concentration*) is the weighted average of the Herfindahl indices of all supplier (customer) industries. The symbol *** indicates statistical significance at the 1% level.

Variables	Mean Market Leverage			Mean Book Leverage			N
	Above median	Below median	Difference (t-value)	Above median	Below median	Difference (t-value)	
Supplier Industries R&D	0.143	0.231	-0.089*** (-66.42)	0.226	0.311	-0.085*** (-40.92)	76,290
Customer Industries R&D	0.211	0.219	-0.009*** (-3.77)	0.284	0.289	-0.005 (-1.36)	26,300
Key Suppliers R&D	0.141	0.207	-0.065*** (-14.10)	0.222	0.282	-0.060*** (-9.66)	4,122
Key Customers R&D	0.196	0.236	-0.040*** (-7.09)	0.262	0.298	-0.036*** (-7.09)	9,452
Supplier SA and JV Intensity	0.157	0.217	-0.060*** (-43.97)	0.246	0.290	-0.045*** (-21.21)	76,290
Customer SA and JV Intensity	0.183	0.244	-0.061*** (-25.99)	0.265	0.306	-0.042*** (-12.25)	26,300
Supplier SA and JV Dummy	0.135	0.189	-0.054*** (-10.46)	0.211	0.268	-0.057*** (-8.23)	4,122
Customer SA and JV Dummy	0.093	0.172	-0.079*** (-16.28)	0.168	0.238	-0.070*** (-9.82)	9,452
Compensation of Employees	0.176	0.198	-0.021*** (-15.36)	0.253	0.283	-0.031*** (-14.37)	76,290
Supplier Concentration	0.196	0.178	0.019*** (13.80)	0.265	0.271	-0.006*** (-2.77)	76,290
Customer Concentration	0.221	0.209	0.011*** (4.99)	0.288	0.286	0.002 (0.67)	26,300

Table 2: Univariate analysis.

6.2 Tables 3 and 4: Supplier industry-level regressions

Read Table 3:

- The dependent variable is market leverage.
- Supplier Industries R&D is negative and statistically significant.
- Supplier SA and JV Intensity is also negative and statistically significant.
- Supplier Concentration is positive, consistent with debt being used as a bargaining tool against powerful suppliers.
- The interaction between Supplier Concentration and Market Share is negative, meaning firms with high own-market power rely less on debt for bargaining.
- The positive interaction between Supplier Industries R&D and Backward Integration Dummy suggests that vertical integration weakens the negative supplier R&D–leverage relation.

Read Table 4:

- The negative relation between supplier R&D and leverage is concentrated among firms with positive own R&D intensity.
- For zero-R&D firms, supplier R&D is not statistically related to leverage.

- Supplier Concentration is positive and significant only for the positive-R&D subsample.

Economic message. Supplier characteristics matter most when the firm itself appears technologically involved. This supports the interpretation that supplier R&D proxies for relationship-specific investment rather than generic supplier sophistication.

Table 3
Ordinary least squares regressions of market leverage on supplier industry-level variables

The sample period is from 1984 to 2003. The dependent variable is *Market Leverage*, which is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. *Supplier Industries R&D* is the weighted average of the R&D intensities of all supplier industries. *Backward Integration Dummy* is a dummy variable that equals one if the firm has a segment in a supplier industry. *Supplier SA and JV Intensity* is the number of SAs and JVs established between industry firms and firms in supplier industries, divided by the number of firms in the industry. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier Concentration* is the weighted average of the Herfindahl indices of all supplier industries. *Supplier Change in Sales* is a weighted average of the change in sales for supplier industries. *Market Share* is the ratio of the firm's sales to total sales of its primary industry. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The reported *t*-values reflect White's heteroskedasticity correction. The symbols ** and *** indicate statistical significance at the 5% and 1% levels, respectively.

	Model 1		Model 2		Model 3	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
Intercept	0.183***	25.66	0.184***	25.59	0.181***	24.86
Supplier Industries R&D	-1.472***	-13.99			-1.208***	-10.48
Dummy Backward Integration					0.786***	3.90
Supplier SA and JV Intensity			-0.011***	-12.09	-0.009***	-9.03
Compensation of Employees	-0.024***	-2.70	-0.025***	-2.79	-0.031***	-3.41
Comp. Of Employees*R&D Intensity					-0.161***	-6.85
Supplier Concentration	0.074***	3.00	0.035	1.42	0.085***	3.14
Supplier Concentration*Market Share					-0.354***	-3.66
Suppliers Change in Sales	-0.102***	-6.40	-0.105***	-6.58	-0.098***	-6.14

Table 3, part 1: OLS regressions on supplier industry-level variables.

Table 3 (continued)

	Model 1		Model 2		Model 3	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
Backward Integration Dummy					0.002	0.66
Market Share					-0.026**	-2.07
Other Control Variables						
Industry Concentration	0.009**	2.24	0.011***	2.69	0.020***	4.62
Firm Size	0.004***	12.71	0.003***	14.25	0.007***	15.73
Asset Collateral Value	0.193***	47.31	0.196***	48.61	0.191***	46.80
Volatility	-0.008***	-4.06	-0.008***	-4.00	-0.006***	-2.71
Nondebt Tax Shield	-4.435***	-18.83	-4.451***	-18.86	-4.364***	-18.54
Return on Assets	-0.110***	-37.96	-0.111***	-38.48	-0.109***	-37.71
Tobin's q	-0.015***	-56.08	-0.015***	-56.20	-0.015***	-55.61
R&D Intensity	-0.232***	-34.63	-0.246***	-36.95	-0.188***	-21.33
SE Intensity	-0.032***	-15.14	-0.031***	-14.76	-0.029***	-13.69
Year and Industry Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	76,290	76,290	76,290	76,290	76,290	76,290
Adjusted R-squared	25.57%		25.54%		26.01%	

Table 3, part 2: Continued supplier industry-level regressions.

Table 4
Ordinary least squares regressions of market leverage on supplier industry-level variables: R&D-based subsamples

The sample period is from 1984 to 2003. The dependent variable is *Market Leverage*, which is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. Column 1 (2) shows OLS regression results for firms with positive (zero) *R&D Intensity*, which is the firm's R&D expenditures divided by its total assets. *Supplier Industries R&D* is the weighted average of the R&D intensities of all supplier industries. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier Concentration* is the weighted average of the Herfindahl indices of all supplier industries. *Supplier Change in Sales* is a weighted average of the change in sales for supplier industries. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The reported *t*-values reflect White's heteroskedasticity correction. The symbols *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	Firms with Positive R&D intensity		Firms with Zero R&D intensity	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
Intercept	0.179***	18.30	0.146***	13.62
Supplier Industries R&D	-1.395***	-12.11	-0.08	-0.26
Compensation of Employees	-0.047***	-3.42	-0.021*	-1.83
Supplier Concentration	0.083***	2.56	-0.001	-0.04
Suppliers Change in Sales	-0.099***	-4.60	-0.117***	-5.17
Other control variables				
Industry Concentration	0.007	1.46	0.016**	2.48
Firm Size	0.003***	6.94	0.009***	15.52
Asset Collateral Value	0.192***	32.08	0.185***	33.77
Volatility	0.001	0.58	-0.016***	-4.21
Nondebt Tax Shield	-3.768***	-16.21	-4.660***	-8.76
Return on Assets	-0.086***	-27.21	-0.140***	-25.05
Tobin's q	-0.011***	-43.41	-0.023***	-33.55
R&D Intensity	-0.199***	-27.22	—	—
SE Intensity	-0.012***	-5.12	-0.050***	-12.67
Year and Industry Dummies	Yes	Yes	Yes	Yes
Number of observations	40,685		35,605	
Adjusted R-squared	22.12%		20.42%	

Table 4: Supplier industry-level regressions for positive- and zero-R&D firms.

6.3 Tables 5 and 6: Customer industry-level regressions

Read Table 5:

- Customer Industries R&D is negatively related to leverage.

- Customer SA and JV Intensity is negatively related to leverage.
- The interactions of customer R&D and customer SA/JV intensity with Market Share are negative, indicating that the effect is stronger when the firm is important to its customers.
- Customer Concentration is positive, supporting the bargaining role of debt with powerful buyers.
- The positive interaction between Customer SA/JV Intensity and Forward Integration Dummy suggests that vertical integration weakens the need to use low leverage as a commitment device.

Read Table 6:

- Customer R&D does not significantly explain leverage for agriculture and mining firms, where relationship-specific investments with customers are less likely.
- The customer R&D effect is negative in durable manufacturing.
- Among positive-R&D firms, customer R&D is strongly negative, and the effect is stronger when the firm's industry is concentrated or when the firm has high market share.

Economic message. Customer-side evidence is especially strong when customers plausibly rely on the firm as a specialized supplier. The customer channel is not merely a general downstream-growth story.

Table 5
Ordinary least squares regressions of market leverage on customer industry-level variables
The sample period is from 1984 to 2003. The dependent variable is *Market Leverage*, which is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. *Customer Industries R&D* is the weighted average of the R&D intensities of all customer industries. *Customer SA and JV Intensity* is the number of SAs and JVs established between industry firms and firms in customer industries, divided by the number of firms in the industry. *Forward Integration Dummy* is a dummy variable that equals one if the firm has a segment in a customer industry. *Customer Concentration* is the weighted average of the Herfindahl indices of all customer industries. *Customer Change in Sales* is a weighted average of the change in sales for customer industries. *Market Share* is the ratio of the firm's sales to total sales of its primary industry. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shield* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The reported *t*-values reflect White's heteroskedasticity correction. The symbols ** and *** indicate statistical significance at the 5% and 1% levels, respectively.

	Model 1		Model 2		Model 3	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
Intercept	0.119***	13.13	0.123***	13.62	0.118***	12.79
Customer Industries R&D	-0.832***	-5.97			-0.875***	-6.38
Customer Industries R&D*Market Share	-1.442***	-3.39				
Customer SA and JV Intensity	0.043	0.30				
Customer SA and JV Intensity*Market Share			-0.015***	-9.18		
Customer SA and JV Intensity*Forward Integration Dummy			-0.070***	-7.43		
Customer Concentration			0.012***	5.69		
Customer Concentration*Market Share	0.101***	3.92	0.033	1.24	0.114***	4.01
Customer Change in Sales	-0.171***	-8.45	-0.168***	-8.33	-0.198	-1.59
Forward Integration Dummy	0.008**	2.26	0.004	1.51	-0.171***	-8.48
Market Share	-0.047***	-4.10	-0.048***	-4.62	-0.026	-1.03

Table 5, part 1: OLS regressions on customer industry-level variables.

Table 5 (continued)

	Model 1		Model 2		Model 3	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
<i>Other control variables</i>						
Industry Concentration	0.044***	5.47	0.050***	6.22	0.044***	5.43
Firm Size	0.006***	8.39	0.007***	9.69	0.006***	8.98
Asset Collateral Value	0.187***	28.72	0.183***	28.29	0.187***	28.70
Volatility	-0.006	-1.31	-0.006	-1.18	-0.006	-1.33
Nondebt Tax Shield	-3.663***	-8.29	-3.432***	-7.80	-3.661***	-8.30
Return on Assets	-0.134***	-21.89	-0.133***	-21.69	-0.135***	-21.98
Tobin's q	-0.019***	-29.68	-0.018***	-29.37	-0.019***	-29.69
R&D Intensity	-0.264***	-15.76	-0.251***	-14.92	-0.265***	-15.85
SE Intensity	-0.044***	-9.25	-0.043***	-9.08	-0.044***	-9.18
Year and Industry Dummies	Yes		Yes		Yes	
Number of observations	26,300		26,300		26,300	
Adjusted R-squared	21.21%		21.40%		21.17%	

Table 5, part 2: Continued customer industry-level regressions.

Table 6
Ordinary least squares regressions of market leverage on customer industry-level variables: various subsamples
The sample period is from 1984 to 2003. The dependent variable is *Market Leverage*. Column 1 show OLS regression results for firms with historical primary SIC code between 0100 and 1499, while Column 2 show OLS regression results for firms producing durable goods. The last three columns show regression results for firms with positive *R&D Intensity*, which is the firm's R&D expenditures divided by its total assets. *Customer Industries R&D* is the weighted average of the R&D intensities of all customer industries. *Customer Concentration* is the weighted average of the Herfindahl indices of all customer industries. *Customer Change in Sales* is a weighted average of the change in sales for customer industries. *Market Share* is the ratio of the firm's sales to total sales of its primary industry. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The *t*-values reported in parentheses reflect White's heteroskedasticity correction. The symbols ** and *** indicate statistical significance at the 5%, and 1% levels, respectively.

	Agriculture and mining	Durable manufacturing	Positive R&D firms	Positive R&D firms	Positive R&D firms
Intercept	0.141*** (3.58)	0.288*** (9.00)	0.179*** (13.46)	0.168*** (12.15)	0.169*** (12.42)
Customer Industries R&D	0.982 (0.95)	-0.841*** (-5.29)	-1.122*** (-6.94)	-0.807*** (-4.24)	-1.101*** (-6.11)
Customer Industries R&D*Industry Concentration					-1.721*** (-3.21)
Customer Industries R&D*Market Share					-2.362*** (-3.36)
Customer Concentration	0.219*** (2.66)	0.184*** (5.19)	0.137*** (3.62)	0.160*** (4.11)	0.164*** (4.23)
Customer Change in Sales	-0.315*** (-5.43)	-0.041 (-1.24)	-0.118*** (-3.70)	-0.114*** (-3.58)	-0.117*** (-3.67)
Industry Concentration	-0.007 (-0.28)		0.015 (1.58)	0.048*** (3.03)	0.025** (2.08)
Market Share		-0.005			0.014 (0.87)

Table 6, part 1: Customer regressions across subsamples.

Other Control Variables	(-0.42)			
Firm Size	0.020*** (4.27)	0.00 (1.57)	-0.001 (-0.93)	-0.001 (-0.91)
Asset Collateral Value	0.199*** (14.25)	0.09*** (7.22)	0.195*** (18.26)	0.194*** (18.27)
Volatility	0.018 (1.43)	-0.03*** (-3.27)	-0.005 (-0.87)	-0.004 (-0.79)
Nondebt Tax Shield	-0.870*** (-4.33)	-2.15*** (-3.40)	-3.555*** (-7.50)	-3.616*** (-7.63)
Return on Assets	-0.08*** (-5.41)	-0.19*** (-15.83)	-0.120*** (-15.38)	-0.120*** (-15.47)
Tobin's q	-0.015*** (-8.44)	-0.02*** (-15.35)	-0.015*** (-21.29)	-0.015*** (-21.19)
R&D Intensity	-0.117 (-1.53)	-0.35*** (-9.80)	-0.248*** (-12.77)	-0.250*** (-12.82)
SE Intensity	-0.073*** (-3.50)	-0.04*** (-2.78)	-0.033*** (-5.34)	-0.033*** (-5.29)
Year and Industry Dummies	Yes	Yes	Yes	Yes
Number of observations	4,993	9,046	10,883	10,883
Adjusted <i>R</i> -squared	18.71%	20.06%	23.27%	23.54%

Table 6, part 2: Continued customer-subsample regressions.

6.4 Tables 7 and 8: Economic significance and robustness

Read Table 7:

- Moving Supplier Industries R&D from the 5th to the 95th percentile lowers predicted leverage by about 0.028, or 14.05 percent.
- Moving Customer Industries R&D from the 5th to the 95th percentile lowers predicted leverage by about 0.054, or 23.59 percent.
- Customer SA and JV Intensity has a similar effect, about 0.056, or 25.31 percent.
- Customer concentration raises predicted leverage, consistent with bargaining.
- In positive-R&D subsamples, the customer R&D effect becomes especially large.

Read Table 8:

- Between-firm regressions continue to show negative R&D effects and positive concentration effects.
- A specification that includes both supplier and customer variables simultaneously gives results similar to the baseline regressions.
- This reduces the concern that supplier variables merely proxy for customer variables, or vice versa.

Economic message. The vertical-chain variables are not small add-ons. Their economic magnitudes are comparable to standard determinants of leverage, and the main patterns survive alternative specifications.

Table 7
Economic significance of regression results
The table reports the predicted values of *Market Leverage* at the 5th and 95th percentiles of each of the independent variables, holding all other independent variables at their sample means. For models with interaction variables, the overall effect for each variable by itself (the *Interaction Effects* column is set to "None") is computed by using the mean of the interaction variable(s). For variables with interaction effects, the predicted values of leverage are also computed at the 5th and 95th percentiles of the interaction variable. *Change* is the difference between the predicted value at the 95th percentile and that at the 5th percentile. The last two columns show the number of the table that contains the model used to compute the predicted values and the number of the model in the respective table. The sample period is from 1984 to 2003. *Supplier (Customer) Industries R&D* is the weighted average of the R&D intensities of all supplier (customer) industries. *Supplier (Customer) SA and JV Intensity* is the number of SAs and JVs established between industry firms and firms in supplier (customer) industries, divided by the number of firms in the industry. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier (Customer) Concentration* is the weighted average of the Herfindahl indices of all supplier (customer) industries. *Supplier (Customer) Change in Sales* is a weighted average of the change in sales for supplier (customer) industries. *Market Share* is the ratio of the firm's sales to total sales of its primary industry. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets.

Independent variable	Interaction effects	Predicted Leverage at the 5th and 95th Percentiles of Independent Variable				Table	Model
		5th perc.	95th perc.	Change	% Change		
Supplier variables							
Supplier Industries R&D	None	0.197	0.169	-0.028	-14.05	3	3
Supplier Industries R&D	Vertically Integrated Firms	0.191	0.181	-0.010	-5.07	3	3
Supplier SA and JV Int.	None	0.192	0.171	-0.016	-8.23	3	3
Supplier Concentration	None	0.184	0.192	0.008	4.17	3	3
Supplier Concentration	5th perc. of Market Share	0.187	0.197	0.010	5.39	3	3
Supplier Concentration	95th perc. of Market Share	0.174	0.171	-0.003	-1.73	3	3
Comp. of Employees	None	0.194	0.181	-0.013	-6.70	3	3
Comp. of Employees	5th perc. of R&D Intensity	0.209	0.199	-0.010	-4.77	3	3
Comp. of Employees	95th perc. of R&D Intensity	0.140	0.116	-0.024	-17.20	3	3
Sup. Change in Sales	None	0.199	0.177	-0.022	-11.27	3	3

Table 7, part 1: Economic significance of regression results.

<i>Customer variables</i>							
Customer Industries R&D	None	0.229	0.175	-0.054	-23.59	5	1
Customer Industries R&D	5th perc. of Market Share	0.232	0.184	-0.049	-20.95	5	1
Customer Industries R&D	95th perc. of Market Share	0.215	0.138	-0.078	-36.18	5	1
Customer SA and JV Int.	None	0.223	0.167	-0.056	-25.31	5	2
Customer SA and JV Int.	Vertically Integrated Firms	0.226	0.201	-0.025	-11.04	5	2
Customer SA and JV Int.	5th perc. of Market Share	0.225	0.184	-0.041	-18.36	5	2
Customer SA and JV Int.	95th perc. of Market Share	0.210	0.087	-0.123	-58.49	5	2
Customer Concentration	None	0.205	0.228	0.023	11.44	5	1
<i>Positive-R&D samples</i>							
Supplier Industries R&D	None	0.151	0.109	-0.042	-27.87	4	1
Customer Industries R&D	None	0.201	0.118	-0.083	-41.31	6	4
Customer Industries R&D	5th perc. of Industry Conc.	0.194	0.131	-0.063	-32.68	6	4
Customer Industries R&D	95th perc. of Industry Conc.	0.093	0.216	-0.123	-56.87	6	4
Customer Industries R&D	5th perc. of Market Share	0.201	0.130	-0.071	-35.22	6	5
Customer Industries R&D	95th perc. of Market Share	0.204	0.066	-0.138	-67.73	6	5
<i>Control variables</i>							
Market Share	None	0.191	0.173	-0.018	-9.50	3	3
Industry Concentration	None	0.184	0.194	0.010	5.54	3	3
Firm Size	None	0.164	0.214	0.051	30.83	3	3
Asset Collateral Value	None	0.137	0.279	0.142	103.09	3	3
Volatility	None	0.188	0.185	-0.004	-2.05	3	3
Nondebt Tax Shield	None	0.191	0.168	-0.022	-11.59	3	3
Return on Assets	None	0.250	0.158	-0.091	-36.63	3	3
Tobin's q	None	0.210	0.124	-0.086	-40.77	3	3
R&D Intensity	None	0.198	0.147	-0.051	-25.81	3	3
SE Intensity	None	0.197	0.169	-0.028	-14.13	3	3

Table 7, part 2: Continued economic significance estimates.

Table 8
Ordinary least squares and between-firm regressions of leverage on supplier and customer industry-level variables
The sample period is from 1984 to 2003. The dependent variable is *Market Leverage*, which is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. Between-firm regressions represent OLS regressions of the mean of the firm's market leverage ratios on the means of independent variables. *Supplier (Customer) Industries R&D* is the weighted average of the R&D intensities of all supplier (customer) industries. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier (Customer) Concentration* is the weighted average of the Herfindahl indices of all supplier (customer) industries. *Supplier (Customer) Change in Sales* is a weighted average of the change in sales for supplier (customer) industries. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The reported *t*-values in the OLS regressions reflect White's heteroskedasticity correction. The symbols *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	Between-firm regressions		Between-firm regressions		OLS Regressions	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
Intercept	0.114***	4.01	0.087**	2.13	0.177***	23.5
<i>Supplier variables</i>						
Supplier Industries R&D	-1.94***	-6.55			-1.01***	-7.94
Compensation of Employees	0.015	0.67			-0.025**	-2.54
Compensation of Employees+R&D Intensity	-0.18**	-2.32			-0.17**	-7.24
Supplier Concentration	0.092*	1.87			0.048*	1.86
Supplier Change in Sales	-0.13**	-2.19			-0.08**	-4.63
<i>Customer variables</i>						
Customer Industries R&D			-0.885**	-2.24	-0.329***	-6.15
Customer Concentration			0.220***	3.11	0.126***	5.72
Customer Change in Sales			-0.164**	-2.26	-0.083***	-4.59

Table 8, part 1: Between-firm and combined supplier/customer regressions.

	Between-firm regressions		Between-firm regressions		OLS Regressions	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
<i>Other control variables</i>						
Industry Concentration	0.004	0.39	0.030*	1.90	0.00	0.99
Firm Size	0.01***	6.81	0.007**	4.83	0.004***	12.17
Asset Collateral Value	0.222***	34.60	0.205***	13.98	0.199***	46.91
Volatility	-0.003	-0.68	-0.010	-1.32	-0.007***	-3.47
Nondebt Tax Shield	-5.819***	-7.63	-5.841***	-4.67	-4.35***	-18.53
Return on Assets	-0.10***	-14.95	-0.118***	-9.42	-0.100***	-37.47
Tobin's q	-0.016***	-22.10	-0.019***	-13.55	-0.011***	-55.92
R&D Intensity	-0.11***	-5.63	-0.265***	-7.66	-0.139***	-21.55
SE Intensity	-0.024**	-4.52	-0.033***	-3.32	-0.03***	-15.15
Year and Industry Dummies	Yes		Yes		Yes	
Number of observations	10,310		4,065		76,160	
Adjusted R-squared	24.39%		19.77%		25.51%	

Table 8, part 2: Continued robustness regressions.

6.5 Tables 9 and 10: Firm-level key supplier evidence

Read Table 9:

- Key Suppliers R&D is negatively related to market leverage.
- The Supplier SA and JV Dummy is also negative and significant.
- The negative key supplier effects are strongest for firms with positive own R&D intensity.
- The firm-level evidence supports the industry-level results using actual key supplier links.

Read Table 10:

- The simultaneous-equation model treats supplier R&D and customer leverage as endogenous.
- Supplier R&D is negatively related to customer market leverage.
- Customer market leverage is negatively related to supplier R&D.
- This supports the idea that leverage and supplier investment decisions are jointly determined.

Economic message. When looking at actual key suppliers rather than industry aggregates, the relationship-specific investment channel still appears. High leverage discourages supplier R&D, and supplier R&D is associated with lower leverage.

Table 9

Ordinary least squares regressions of market leverage on key supplier variables

The sample period is from 1984 to 2002. The dependent variable is *Market Leverage*, which is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. Column 2 (3) show OLS regression results for firms with positive (zero) *R&D Intensity*, which is the firm's R&D expenditures divided by its total assets. *Key Suppliers R&D* is the weighted average of the R&D intensities of key suppliers. *Suppliers SA and JV Dummy* is a dummy variable that equals one if the firm has at least one alliance or a SA established with one of its suppliers, and zero otherwise. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier Concentration* is the weighted average of the Herfindahl indices of all supplier industries. *Supplier Change in Sales* is a weighted average of the change in sales for supplier industries. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The reported *t*-values reflect White's heteroskedasticity correction. The symbols *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	All customers		Positive R&D customers		Zero R&D customers	
	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value	Coeff.	<i>t</i> -value
Intercept	0.237***	6.71	0.242***	5.17	0.217***	3.48
Key Suppliers R&D	-1.632***	-5.68	-1.687***	-6.03	-1.070	-1.35
Suppliers SA and JV Dummy	-0.014***	-3.15	-0.038***	-5.62	0.046	1.63
<i>Control variables</i>						
Compensation of Employees	-0.135***	-4.09	-0.187***	-5.32	-0.013	-0.18
Supplier Concentration	0.027	0.37	0.220**	2.47	-0.395***	-2.95
Supplier Change in Sales	-0.070	-1.59	-0.068	-1.26	-0.068	-0.89
Industry Concentration	0.000	-0.34	-0.030*	-1.95	0.066*	1.72
Firm Size	0.002	1.21	0.006***	3.59	-0.002	-0.67
Asset Collateral Value	0.088***	5.38	0.07***	3.29	0.140***	5.01
Volatility	-0.014	-0.67	-0.013	-0.46	-0.017	-0.36
Nondebt Tax Shield	-4.653***	-5.37	-2.751***	-3.24	-11.615***	-3.64
Return on Assets	-0.282***	-12.41	-0.30***	-12.74	-0.262	-5.04
Tobin's q	-0.023***	-14.03	-0.015***	-11.27	-0.029***	-6.63
R&D Intensity	-0.336***	-8.31	-0.337***	-8.26	—	—
SE Intensity	-0.101***	-8.15	-0.09***	-7.62	-0.089**	-2.08
Year and Industry Dummies	Yes		Yes		Yes	
Number of observations	4,122		2,932		1,190	
Adjusted R-squared	41.01%		40.31%		36.48%	

Table 9: OLS regressions on key supplier variables.

Simultaneous equations: supplier R&D and leverage of main customer

This table reports the second stage estimates of a 2SLS estimation of a simultaneous equation model that treats the supplier R&D and the leverage of its main customer as endogenous variables. The sample period is from 1984 to 2002. *Customer Market Leverage* is the market leverage of the customer that buys the highest percentage of the supplier's output, where market leverage is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of book value of debt and market value of common equity. *Supplier R&D* is supplier's R&D expenditures divided by total assets. *Compensation of Employees* is total compensation paid to employees divided by total output. *Supplier Concentration* is the weighted average of the Herfindahl indices of all supplier industries. *Supplier Change in Sales* is a weighted average of the change in sales for supplier industries. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. *Supplier Size* is the log of the supplier's total assets. *Supplier Book-to-Market* is the supplier's total assets divided by the sum of the book value of debt and the market value of common equity. *Supplier Surplus Cash* is the supplier's cash from assets-in-place divided by total assets. *Supplier Sales Growth* is the supplier's one-year growth in total sales. *Supplier Stock Return* is the supplier's one-year stock return. *Supplier Leverage* is the supplier's book leverage. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The symbols *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	Customer market leverage		Supplier R&D	
	Coefficient	<i>t</i> -value	Coefficient	<i>t</i> -value
<i>Dependent variable</i>				
Intercept	0.277***	5.93	0.174***	5.11
Supplier R&D	-0.164**	-2.13		
Customer Market Leverage			-0.202***	-4.95
<i>Control variables</i>				
Compensation of Employees	-0.129***	-3.47		
Supplier Concentration	0.049	0.49		
Supplier Change in Sales	-0.076	-1.23		
Industry Concentration	-0.007	-0.35		
Firm Size	-0.002	-0.90		
Asset Collateral Value	0.039*	1.93		
Volatility	-0.031	-1.22		
Nondebt Tax Shield	-4.903***	-3.47		
Return on Assets	-0.248***	-10.89		
Tobin's q	-0.017***	-9.05		
R&D Intensity	-0.333***	-5.66		
SE Intensity	-0.114***	-8.23		
Supplier Size			-0.017***	-12.07
Supplier Book-to-Market			-0.084***	-11.19
Supplier Surplus Cash			0.026**	2.07
Supplier Sales Growth			0.003	1.19
Supplier Stock Return			-0.018***	-6.34
Supplier Leverage			-0.010***	-2.98
Year Industry Dummy Variables	Yes		Yes	
Number of observations	2,185		2,185	
Adjusted R-squared	38.95%		27.27%	

Table 10: Simultaneous equations for supplier R&D and main-customer leverage.

6.6 Tables 11 and 12: Firm-level key customer evidence

Read Table 11:

- Key Customers R&D is negatively related to market leverage in the full sample.
- The Customer SA and JV Dummy is negative and significant.
- The effects are stronger for positive-R&D suppliers and for durable manufacturing firms.
- The relation is not significant for zero-R&D suppliers and agriculture/mining firms, which is consistent with the relationship-specific investment interpretation.

Read Table 12:

- The simultaneous-equation model treats customer R&D and supplier leverage as endogenous.
- Customer R&D is negatively related to supplier market leverage.
- Supplier market leverage is negatively related to customer R&D.
- The results mirror the supplier-side simultaneous-equation evidence.

Economic message. The key-customer results complete the two-sided evidence: both suppliers and customers appear less willing to make R&D investments when the focal firm is highly levered.

Table 11
Ordinary least squares regressions of market leverage on key customer variables
The sample period is from 1984 to 2002. The dependent variable is *Market Leverage*, which is the sum of the book values of long-term debt and debt in current liabilities divided by the sum of the book value of debt and the market value of common equity. Column 2 (3) show OLS regression results for firms with positive (zero) *R&D Intensity*, which is the firm's R&D expenditures divided by its total assets. Column 4 show OLS regression results for firms with historical primary SIC code between 0100 and 1499. Column 5 show OLS regression results for firms producing durable goods, based on the Census Bureau classification. *Key Customer R&D* is the weighted average of the R&D intensities of key customers. *Customers SA and JV Dummy* is a dummy variable that equals one if the firm has at least one alliance or a SA established with one of its customers, and zero otherwise. *Customer Concentration* is the weighted average of the Herfindahl indices of all supplier industries. *Customer Change in Sales* is a weighted average of the change in sales for supplier industries. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The *t*-values reported in parentheses reflect White's heteroskedasticity correction. The symbols *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	All suppliers	Positive R&D suppliers	Zero R&D suppliers	Agriculture and mining	Durable manufacturing
Intercept	0.006*** (5.63)	0.06*** (3.41)	0.127*** (4.32)	0.093 (1.52)	0.206*** (3.56)
Key Customers R&D	-0.209*** (-2.86)	-0.209*** (-2.78)	0.108 (0.47)	0.985 (0.70)	-0.226** (-2.00)
Customer SA and JV Dummy	-0.029*** (-8.37)	-0.02** (-5.79)	-0.019 (-1.52)	-0.059* (-1.83)	-0.019*** (-3.87)
<i>Control variables</i>					
Customer Concentration	0.145*** (3.02)	0.211*** (3.55)	-0.133 (-1.51)	-0.233 (-1.35)	0.147*** (2.63)
Customer Change in Sales	-0.115** (-2.14)	-0.16** (-2.54)	-0.082 (-1.04)	0.009 (0.07)	-0.167** (-2.02)
Industry Concentration	0.022*** (2.73)	0.018 (1.38)	0.081*** (3.55)	0.090 (1.37)	0.022*** (2.13)
Firm Size	0.007*** (6.76)	0.003*** (3.54)	0.017*** (8.65)	0.022*** (5.63)	0.002 (1.32)

Table 11, part 1: OLS regressions on key customer variables.

	All suppliers	Positive R&D suppliers	Zero R&D suppliers	Agriculture and mining	Durable manufacturing
Asset Collateral Value	0.246*** (21.07)	0.254*** (19.11)	0.203*** (9.90)	0.140*** (3.04)	0.260*** (14.67)
Volatility	0.000 (-0.06)	0.00 (-0.44)	0.013 (0.85)	-0.038 (-0.89)	-0.014* (-1.75)
Nondebt Tax Shield	-5.044*** (-9.32)	-4.380*** (-9.15)	-4.380*** (-9.34)	-5.639 (-1.48)	-4.422*** (-6.71)
Return on Assets	-0.105*** (-13.36)	-0.099*** (-11.32)	-0.133*** (-6.68)	-0.069 (-1.60)	-0.139*** (-11.82)
Tobin's q	-0.014*** (-22.03)	-0.01*** (-19.47)	-0.024*** (-9.68)	-0.023* (-1.85)	-0.015*** (-14.69)
R&D Intensity	-0.197*** (-12.22)	-0.14*** (-8.84)	—	0.133 (0.50)	-0.325*** (-9.60)
SE Intensity	-0.001 (-0.21)	-0.003 (-0.40)	-0.013 (-0.79)	-0.035 (-0.51)	0.097 (0.52)
Year and Industry Dummies	Yes	Yes	Yes	Yes	Yes
Number of observations	9,452	6,175	3,277	9,07	4,645
Adjusted R-squared	31.28%	30.38%	24.86%	16.33%	30.20%

Table 11, part 2: Continued key customer regressions.

Simultaneous equations (2SLS): customer R&D and leverage of main supplier

This table reports the second stage estimates of a 2SLS estimation of a simultaneous equation model that treats the customer R&D and the leverage of its main supplier as endogenous variables. The sample period is from 1984 to 2002. *Supplier Market Leverage* is the market leverage of the main supplier, where market leverage is the book value of long-term debt and debt in current liabilities divided by the sum of book value of debt and market value of common equity. *Customer R&D* is customer's R&D expenditures divided by total assets. *Customer Concentration* is the weighted average of the Herfindahl indices of all customer industries. *Customer Change in Sales* is a weighted average of the change in sales for customer industries. *Industry concentration* is the sales-based Herfindahl index of the firm's industry. *Firm Size* is the log of total assets. *Asset Collateral Value* is net property, plant, and equipment divided by total assets. *Volatility* is the standard deviation of operation income divided by total assets. *Nondebt Tax Shields* is investment tax credits divided by total assets. *Return on Assets* is operation income divided by total assets. *Tobin's q* is book value of assets minus book value of common equity plus market value of common equity divided by book value of assets. *R&D Intensity* is R&D expenditures divided by total assets. *SE Intensity* is selling, general, and administrative expenses divided by total assets. *Customer Size* is the log of the customer's total assets. *Customer Book-to-Market* is the customer's total assets divided by the sum of the book value of debt and market value of common equity. *Customer Surplus Cash* is the customer's cash from assets-in-place divided by total assets. *Customer Sales Growth* is the customer one-year growth in total sales. *Customer Stock Return* is the customer one-year stock return. *Customer Leverage* is the supplier book leverage. All variables are winsorized at the 1st and 99th percentiles. Lagged values of firm-specific control variables are used. The symbols ** and *** indicate statistical significance at the 5% and 1% levels, respectively.

	Supplier Market Leverage		Customer R&D	
	Coefficient	<i>t</i> -value	Coefficient	<i>t</i> -value
<i>Dependent Variable</i>				
Intercept	0.328***	4.87	0.258***	5.39
Customer R&D	-0.585**	-2.16		
Supplier Market Leverage			-0.258***	-2.86
<i>Control variables</i>				
Customer Concentration	0.234	1.51		
Customer Change in Sales	-0.207	-1.12		
Industry Concentration	-0.005	-0.08		
Firm Size	-0.016	-0.64		
Asset Collateral Value	0.164***	3.66		
Volatility	-0.113***	-2.81		
Nondebt Tax Shield	-9.412**	-2.18		
Return on Assets	-0.113**	-2.48		
Tobin's q	-0.013***	-2.96		
R&D Intensity	0.172	1.00		
SE Intensity	-0.111**	-2.18		
Customer Size			-0.015***	-4.35
Customer Book-to-Market			-0.108***	-5.10
Customer Surplus Cash			0.035	1.02
Customer Sales Growth			0.005	0.67
Customer Stock Return			-0.014	-1.50
Customer Leverage			-0.022	-0.73
Year and Industry Dummy Variables	Yes		Yes	
Number of observations	385		385	
Adjusted R-squared	27.05%		28.65%	

Table 12: Simultaneous equations for customer R&D and main-supplier leverage.

6.7 Appendix tables: final users and supplier-customer pairs

Read:

- Table B.1 shows examples of industries that sell different shares of output to final users versus intermediate users. This motivates the customer-sample restrictions.
- Table C.1 shows examples of supplier-customer industry pairs identified from Compustat segment files, such as motor vehicle parts suppliers and motor vehicle manufacturers.
- These appendix tables help clarify how the paper constructs vertical-chain relationships rather than assuming all firms have identifiable corporate customers.

Economic message. The appendix reinforces that customer identification is nontrivial. Some industries mainly sell to final users, while others sell almost entirely into intermediate production chains.

Table B.1

Industry name	SIC Code	% used as intermediate input	% sold to final users	Main customer industry/ final users	Another significant customer industry/final users
Motor Homes	3716	0	100	Personal Consumption Expenditures	N/A
Small Arms Ammunition	3482	1	99	Federal Government	N/A
Motor Vehicles and Passenger Car Bodies	3711	2	98	Personal Consumption Expenditures	N/A
Petroleum Refining	2911	52	48	Personal Consumption Expenditures	Air Transportation
Aircraft Engines and Engine Parts	3724	63	37	Aircraft	Federal Government
Motors and Generators	3621	83	17	Gross Private Investments	Refrigeration and Heating Equipment
Motor Vehicle Parts and Accessories	3714	93	7	Motor Vehicles and Passenger Car Bodies	Automotive Repair Shops And Services
Sheet Metal Work	3444	96	4	Motor Vehicle Parts and Accessories	Industrial and Commercial Buildings Construction
Screw Machine Products	3451	98	2	Motor Vehicles and Passenger Car Bodies	Motor Vehicle Parts and Accessories
Nonferrous Foundries (castings)	3360	99	1	Motor Vehicle Parts and Accessories	Industrial and Commercial Machinery and Eqt.
Primary Smelting and Refining of Copper	3331	100	0	Rolling, Drawing, and Extruding of Copper	Industrial and Commercial Machinery and Equipment
Primary Production of Aluminum	3334	100	0	Aluminum Rolling and Drawing	Aluminum Castings
Plastics Materials	2820	100	0	Miscellaneous Plastics Products	Paints and Allied Products

Table B.1: Percentage of industry output sold to final users.

Table C.1

Supplier industry		Customer industry		Number of pairs	Number of unique customers	Example	
Name	SIC code	Name	SIC code			Supplier	Main customer
Motor Vehicle Parts and Accessories	3714	Motor Vehicles and Passenger Car Bodies	3711	16	3	Dana Corp	Ford Motor Co
Biological Products	2836	Pharmaceutical Preparations	2834	12	11	Medarex Inc	Lilly (Eli) & Co
Telephone and Telegraph Apparatus	3661	Telephone Communications	4813	11	8	Telabs Inc	Verizon
Special Industry Machinery	3559	Communications Semiconductors and Related Devices	3674	8	5	Novellus Systems	Intel Corp
Semiconductors and Related Devices	3674	Radio and Television Broadcasting and Communications Eqt.	3663	6	3	Triquint Semiconductor Inc	Motorola Inc
Surgical and Medical Instruments and Apparatus	3841	Pharmaceutical Preparations	2834	5	3	Micro Therapeutics Inc	Abbott Laboratories
Electromedical and Electrotherapeutic Apparatus	3845	Pharmaceutical Preparations	2834	5	5	Healthetech Inc	Natures Sunshine Prods
Instruments for Measuring and Testing of Electricity and Electrical	3825	Semiconductors and Related Devices	3674	5	4	Ltx Corp	Texas Instruments Inc
Semiconductors and Related Devices	3674	Electronic Computers	3571	4	2	Tripath Technology Inc	Apple Computer

Table C.1: Examples of supplier-customer industry pairs.

7 Short summary

- The paper studies whether firms' leverage choices reflect characteristics of suppliers and customers.
- It tests two theories: low leverage as a commitment to trading partners that make relationship-specific investments, and high leverage as a bargaining tool against concentrated trading partners.
- Relationship-specific investment is proxied by supplier/customer R&D intensity and by strategic alliance and joint venture intensity.
- Supplier/customer bargaining power is proxied by concentration in supplier and customer industries.
- Industry-level data are built from U.S. input-output tables and Compustat.
- Firm-level tests use identified key suppliers and customers from Compustat segment disclosures.
- Leverage is negatively related to supplier and customer R&D intensity.
- Leverage is negatively related to supplier/customer alliance and joint venture intensity.

- The negative effects are weaker for vertically integrated firms, consistent with vertical integration substituting for leverage as a contracting mechanism.
- Customer-side effects are stronger when the firm has high market share or operates in a concentrated industry, consistent with the firm being important to its customers.
- Leverage is positively related to supplier and customer concentration, consistent with debt helping firms bargain with powerful trading partners.
- Simultaneous-equation tests suggest leverage and trading-partner R&D decisions are jointly determined.
- The effects are economically large and comparable to standard capital-structure determinants.

8 Conclusion

8.1 Core conclusion

Kale and Shahrur show that capital structure is shaped by vertical-chain relationships. Firms use lower leverage when suppliers or customers are likely to make relationship-specific investments, and use higher leverage when concentrated suppliers or customers have more bargaining power.

8.2 Economic intuition

Debt has two opposing roles. On one hand, too much debt raises the likelihood of financial distress and weakens the firm's incentive to maintain reputation and long-term relationships. This discourages suppliers and customers from making specialized investments. On the other hand, debt can strengthen the firm's bargaining position by committing cash flows away from powerful trading partners.

8.3 Incremental contribution revisited

The paper's main contribution is to integrate suppliers and customers into empirical capital-structure analysis. It shows that leverage is not only a response to firm-level balance-sheet variables, but also to the structure of the firm's input and output markets.

8.4 Implications

Capital-structure models should account for implicit contracts with outside stakeholders. Firms embedded in technology-intensive vertical chains may rationally carry less debt even if traditional determinants suggest otherwise. Firms facing concentrated suppliers or customers may rationally carry more debt because bargaining considerations matter.

8.5 Limitations

The evidence is observational. R&D intensity, alliances, and JVs are indirect proxies for relationship-specific investment. Industry-level input-output data cannot perfectly identify every firm's actual

suppliers and customers. Firm-level key supplier/customer tests are more precise but smaller and more vulnerable to endogeneity. The simultaneous-equation analysis helps, but it does not fully establish causality.

8.6 Takeaway

The enduring takeaway is that financing policy is part of product-market strategy. Firms choose leverage not only to manage taxes, distress costs, and agency problems, but also to sustain supplier and customer relationships and to bargain within the vertical production chain.